Object Detection in High Resolution Remote Sensing Imagery Based on Convolutional Neural Networks With Suitable Object Scale Features

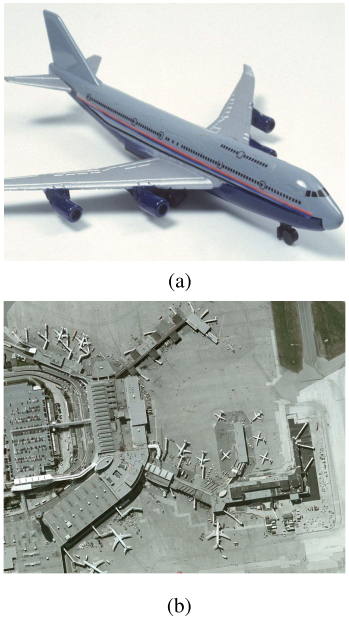
Zhipeng Dong, Mi Wang, Yanli Wang, Ying Zhu, and Zhiqi Zhang

***Abstract*— Object detection in high spatial resolution remote sensing images (HSRIs) is an important part of image information automatic extraction, analysis, and understanding. The region of interest (ROI) scale of object detection and the object feature representation are two vital factors in HSRI object detection. With respect to these two issues, this article presents a novel HSRI object detection method based on convolutional neural networks (CNNs) with suitable object scale features. First, the suitable ROI scale of object detection is obtained by compiling statistics for the scale range of objects in HSRIs. Then, a CNN framework for object detection in HSRIs is designed using a suitable ROI scale of object detection. The object features obtained using a CNN have good universality and robustness. Finally, a CNN framework with a suitable ROI scale of object detection is trained and tested. Using the WHU-RSONE data set, the proposed method is compared with the faster region-based CNN (Faster-RCNN) framework. The experimental results show that the proposed method outperforms the Faster-RCNN framework and provides good object detection results in HSRIs.**

***Index Terms*— Convolutional neural network (CNN), deep learning, high-resolution remote sensing image, object detection, object scare.**

I. Introduction

With the development of Earth observation satellite technology, a large amount of high spatial resolution remote sensing images (HSRIs) can be obtained [1]–[3]. Object detection using HSRIs is a fundamental method for image information automatic extraction, analysis, and understanding, and plays an important role in both civilian and military applications [4]–[10]. Object detection in HSRIs is to determine whether an HSRI contains one or more objects belonging to the class of interest and locate the position of each predicted object in the image [11]. The term "object" used in this article mainly refers to man-made objects (e.g., airplanes, storage tanks, or ships) that have clear boundaries and are independent of the image background. Compared with natural images, HSRIs have the characteristics of more complex image background and smaller object scale, as shown in Fig. 1. Object detection in HSRIs is more challenging than that in natural images [12]. Therefore, the problem of object detection in HSRIs has attracted different scholars' research attention.



(a) (b)

Fig. 1. (a) Natural image. (b) HSRI.

So far, many detectors have been developed to detect different types of objects in remote sensing images [11]. Some researchers used handcrafted features to extract characteristics, such as saliency [13], scale-invariant feature transform (SIFT) [14], histogram of oriented gradients (HOG) [15], and the bag-of-words (BOW) model [16]. Diao *et al.* [17] proposed an efficient coarse object locating method based on a saliency mechanism. After that, a trained deep belief network was used for feature extraction and classification for located regions. Tao *et al.* [18] simulated an airport with a set of SIFT keypoints and detected it using an improved SIFT matching strategy. Xiao *et al.* [19] used orientation normalization, feature space mapping, and an elliptic Fourier transform to achieve rotational invariance of HOG. Aircrafts and cars in remote sensing images can be detected using the rotational invariance of HOG. Sun *et al.* [20] searched for the locations of targets using a sliding window and extracted features of

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targets based on sparse spatial coding BOW (SSCBOW). SSCBOW representation was then combined with a linear support vector machine for target detection. Handcrafted features perform well for specific object detection tasks. However, the universality and robustness of handcrafted features for other object detection tasks are poor [11].

Deep learning models have recently received attention. The most popular deep learning methods are the convolutional neural network (CNN) models [21]–[29]. A CNN does not need handcrafted features and requires fewer parameters than other networks because it can share weights for the same filter. Essential features of input images can be learned using CNN based on its specific network structure [30]. The object features obtained using CNN exhibit good universality and robustness for different object detection tasks [31]. CNNs have been widely used in object detection of remote sensing images. Long *et al.* [30] used a selective search algorithm [32] to generate category-independent possible regions. The candidate regions were sent to a combined model consisting of 2-D reduction CNNs for classification. Finally, the unsupervised score-based bounding box regression method was used to improve box localization precision after using non-maximum suppression (NMS) [33]. Han *et al.* [34] detected various geospatial objects in HSRIs using faster region-based CNN (Faster-RCNN) [24]. In the Faster-RCNN framework, a pretraining mechanism was utilized to improve the efficiency of the various geospatial traits by transfer learning from the natural imagery domain to the HSRI domain. Chen *et al.* [35] used transfer learning and adopted a single deep CNN and limited training samples to implement end-to-end trainable airplane detection. An airplane detection CNN framework was based on single-shot multibox detector (SSD) [26]. Li *et al.* [36] developed a unified object detection CNN framework, which combines well the region proposal network (RPN) and the local-contextual feature network. The proposed framework can significantly improve object detection performance. Deng *et al.* [37] redesigned the feature extractor of CNN by adopting concatenated rectified linear unit (ReLU) and inception modules. Detection is then performed by two subnetworks: a multiscale object proposal network for object-like region generation from several intermediate layers and an accurate object detection network for object detection based on fused feature maps.

In HSRI object detection methods based on CNN, human-labeled training examples and region of interest (ROI) scale of object detection are two vital factors determining the HSRI object detection results.

Human-labeled training examples are "fuel" for supporting the performance of HSRI object detection methods based on CNN. However, the current human-labeled training examples are costly and limited for HSRI object detection methods based on CNN [38].

Although the Faster-RCNN and SSD frameworks have been proven to be successful for detecting objects (e.g., horses, dogs, or people) in natural images, they are not specifically designed to detect small objects in HSRIs. Some challenges when using HSRIs limit their application in the field of earth observation [37]. A good detector should be able to cover most of the object sizes. A few HSRI object detection methods based on CNN consider the ROI scale of object detection in images.

For solving the issue of a lack of human-labeled training examples, we have established an HSRI object detection data set, WHU-RSONE, described in this article. There are 51 866 geospatial objects (15 703 airplanes, 24 692 storage-tanks, and 11 471 ships) in WHU-RSONE.

With respect to the ROI scale of object detection in images, a novel object detection method based on CNNs with suitable object scale features (CNN-SOSFs) is proposed in this article. First, the suitable ROI scale of object detection is obtained by compiling statistics for the scale range of objects in HSRIs. Then, the CNN-SOSF framework is designed using the suitable ROI scale of object detection. The object features obtained using CNN-SOSF have good universality and robustness. Finally, the CNN-SOSF framework is trained and tested using WHU-RSONE.

The main contributions of this article are summarized as follows.

1) An HSRI object detection data set, WHU-RSONE, is established and made publication to support the development of related studies.
2) The suitable ROI scales of object (airplane, storage tank, and ship) detection are obtained by compiling statistics for a large number of objects with different scales in WHU-RSONE.
3) Using suitable ROI scales of object detection, a novel object detection method based on CNN-SOSFs is proposed for the object detection of HSRIs.

The rest of this article is organized as follows. Section II introduces the CNN-SOSF framework and how to train and test it. Section III describes the test data and presents the results and discussion. Section IV summarizes the results and presents conclusions.

II. Methodology

In this article, there are two steps to obtain the CNN-SOSF framework. First, the suitable ROI scale of object is obtained by statistical methods addressing the scale range of objects in HSRIs. Second, a CNN-SOSF framework is designed using the suitable ROI scale of object detection. The method used to train and test the CNN-SOSF framework is described.

A. ROI Scale Determination of Object Detection

HSRIs are obtained by satellite or space sensors in near-Earth orbit using a top–down view, which is an approach that is susceptible to the weather and illumination conditions [34]. There are complex scenes in HSRIs [39], [40]. Furthermore, objects are relatively small in large-scale HSRIs and appear in densely distributed groups [37], [41]. Therefore, the ROI scale of object detection is a vital factor affecting the detection results of HSRI object detection methods based on CNN. A good detector should be able to cover most of object sizes. In this article, an HSRI object detection data set, WHU-RSONE, is established. There are 5977 HSRIs in WHU-RSONE, and the size of images ranges from 600×600 pixels

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TABLE I
NUMBER OF OBJECTS IN WHU-RSONE

Object	Number
Airplane	15,703
Storage-tank	24,692
Ship	11,471
Total	51,866

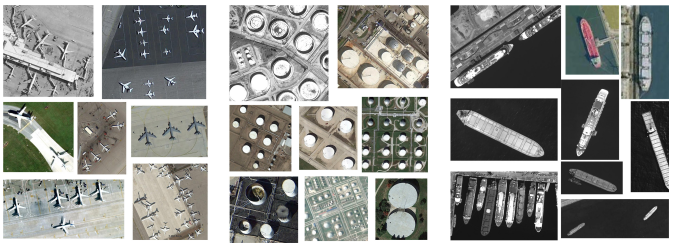


Fig. 2. Object samples in WHU-RSONE.

to 1372×1024 pixels. The number of objects in WHU-RSONE is shown in Table I. In WHU-RSONE, airplanes, storage tanks, and ships have different scales; object samples are shown in Fig. 2. Therefore, the ROI scale of object (airplanes, storage tanks, and ships) detection in HSRIs can be obtained by compiling statistics for a large number of objects with different scales in WHU-RSONE.

Recently, some researchers have begun using CNN to generate object proposal regions [34], [36], [37]. Ren *et al.* [24] proposed an RPN for generating object proposal regions in Faster-RCNN framework, which significantly improves the object proposal regions' quality and the final object detection accuracy because the RPN uses anchors that have the characteristics of multiscale and translation invariant [36]. In the RPN of Faster-RCNN, three scales (128 pixels, 256 pixels, and 512 pixels) and three aspect ratios (1:1, 1:2, and 2:1) are adopted to yield nine anchors at each sliding position. Nine anchors are shown in the blue rectangle of Fig. 3. The anchor scales of RPN of Faster-RCNN are specifically designed for natural scene images. Compared to natural scene images, object scales in HSRIs are smaller, as shown in Fig. 1. Therefore, in our RPN, four scales (16 pixels, 32 pixels, 64 pixels, and 128 pixels) and three aspect ratios (1:1, 1:2, and 2:1) are adopted to yield 12 anchors at each sliding position to form multiscale bounding boxes and 12 anchors as shown in the red rectangle of Fig. 3. The RPN nine anchors' coverage area of Faster-RCNN is shown in the blue area of Fig. 4. Our RPN 12 anchors' coverage area is shown in the red area of Fig. 4.

Using the WHU-RSONE training set, the various scales of airplanes, storage tanks, and ships are compiled statistics. Fig. 5(a)–(c) shows the statistical results for the scales of airplanes, storage tanks, and ships, respectively. Using the WHU-RSONE training set, all object (airplane, storage tank, and ship) scales are compiled statistics together, and the statistical results are shown in Fig. 5(d).

In Fig. 5(a), 6.58% of airplane scales are in the RPN nine anchors' coverage area of Faster-RCNN and 96.68% of airplane scales are in the RPN 12 anchors' coverage area.

In Fig. 5(b), 0.65% of storage-tank scales are in the RPN nine anchors' coverage area of Faster-RCNN and 97.5% storage-tank scales are in the RPN 12 anchors' coverage area.

In Fig. 5(c), 24.24% of ship scales are in the RPN nine anchors' coverage area of Faster-RCNN and 90% ship scales are in the RPN 12 anchors' coverage area.

In Fig. 5(d), 7.62% of all object scales are in the RPN nine anchors' coverage area of Faster-RCNN and 95.61% of all object scales are in the RPN 12 anchors' coverage area.

Statistical results show that our RPN 12 anchors' coverage area can cover object scales better than the RPN nine anchors' coverage area of Faster-RCNN and cover 95.61% of object scales in the WHU-RSONE training set. Our RPN 12 anchors can fit object scales in HSRIs well. ROI of object detection in HSRIs is obtained in this article using RPN with four scales (16 pixels, 32 pixels, 64 pixels, and 128 pixels) and three aspect ratios (1:1, 1:2, and 2:1).

B. CNN-SOSF Framework Design

Using suitable ROI scales for object detection, we propose a novel object detection framework based on CNN-SOSFs for object detection in HSRIs. The CNN-SOSF framework consists of two sequential modules: RPN (generating multiscale- and translation-invariant object proposal regions) and object classification and accurate localization CNN (generating class labels and reducing localization errors for each object proposal region). The CNN-SOSF framework is shown in Fig. 6. Next, each module in the CNN-SOSF framework is described.

1) RPN: An RPN can input any size of image and output a series of proposal regions with object probability scores [24], [36]. We are particularly inspired by the results from the Faster-RCNN, in which multiscale- and translation-invariant object proposal regions are generated using an RPN. Our RPN has been adapted to HSRIs by counting the scale range of objects in HSRIs, and this technique is used to effectively solve the problem of the small scale of HSRI objects. In the RPN of CNN-SOSF, four scales (16 pixels, 32 pixels, 64 pixels, and 128 pixels) and three aspect ratios (1:1, 1:2, and 2:1) are used to yield 12 anchors at each sliding position to form multiscale object proposal regions.

In CNN-SOSF, object proposal regions are generated on the last feature map using RPN. The sliding window with step size one is used in the last feature map. At each sliding window location, k object proposal regions are simultaneously predicted. Then, the reg layer of RPN has $4k$ outputs representing the coordinates of k object proposal regions, as shown in Fig. 3. The cls layer has $2k$ score outputs to estimate the probability of object or not-object for each object proposal region. The parameter k is 12 in this article, which is relative to the number of anchors. The k object proposal regions are the 12 anchors whose centers are at sliding window location. To train an RPN, a binary class label for being an object or not-object is assigned to each anchor. A positive label is

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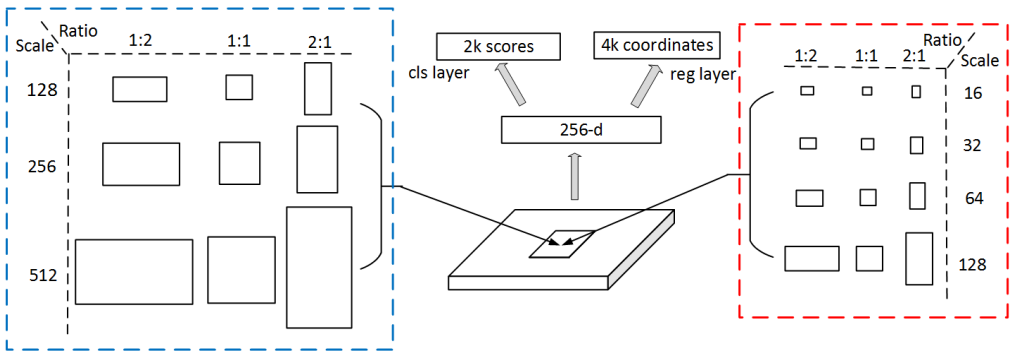


Fig. 3. RPN architecture.

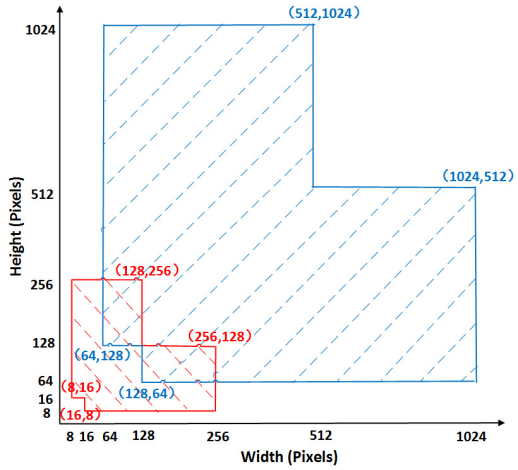


Fig. 4. RPN anchor coverage area.

assigned to two kinds of anchors: 1) the intersection-over-union (IoU) overlap of this anchor and a ground truth box is the greatest among that of all anchors and the ground truth box or 2) the IoU overlap of this anchor and any ground truth box is greater than 0.7. If the IoU overlap of an anchor and any ground-truth boxes is less than 0.3, a negative label is assigned to the anchor. Anchors that are neither positive nor negative are not used for the training objective of RPN. The multitask loss function $L(\{p_i\},\{t_i\})$ of training RPN is calculated as follows:

$$L(\{p_i\},\{t_i\}) = \frac{1}{N_{\text{cls}}}\sum_i L_{\text{cls}}\left(p_i, p_i^*\right) + \lambda \frac{1}{N_{\text{reg}}}\sum_i p_i^* L_{\text{reg}}\left(t_i, t_i^*\right) \tag{1}$$

$$L_{\text{cls}}\left(p_i, p_i^*\right) = -p_i^* ln(p_i) \tag{2}$$

$$L_{\text{reg}}\left(t_i, t_i^*\right) = \sum_{i\in\{x,y,x,h\}} \text{smooth}_{L_1}\left(t_i - t_i^*\right) \tag{3}$$

$$\begin{cases} t_x = \dfrac{x_i - x_a}{w_a} & t_y = \dfrac{y_i - y_a}{h_a} \quad t_w = \ln\dfrac{w_i}{w_a} \quad t_h = \ln\dfrac{h_i}{h_a} \\ t_x^* = \dfrac{x_i^* - x_a}{w_a} & t_y^* = \dfrac{y_i^* - y_a}{h_a} \quad t_w^* = \ln\dfrac{w_i^*}{w_a} \quad t_h^* = \ln\dfrac{h_i^*}{h_a} \end{cases} \tag{4}$$

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise} \end{cases} \tag{5}$$

where i is the index of an anchor in a mini-batch and p_i is the predicted probability of anchor i being an object. If the anchor is positive, the ground-truth label p_i^* is 1; otherwise, it is 0. t_i is a vector representing the four parameterized coordinates of the predicted anchor i and is $[x_i, y_i, w_i, h_i]^T$. t_i^* is a vector representing the four parameterized coordinates of the ground-truth box associated with a positive anchor and is $[x_i^*, y_i^*, w_i^*, h_i^*]^T$. $[x_a, y_a, w_a, h_a]^T$ is the four parameterized coordinates' vector of anchor i. The classification loss L_{cls} is the standard cross-entropy loss over two classes (object or not-object). The regression loss L_{reg} is the smooth L_1 loss function. The two losses are normalized using N_{cls} ($N_{\text{cls}} = 256$), N_{reg} ($N_{\text{reg}} = 2400$), and a balancing weight λ ($\lambda = 10$) [24].

2) Object Classification and Accurate Localization CNN: The object classification and accurate localization CNN generates class labels of object proposal regions and achieves accurate localization of object proposal regions. In the object classification and accurate localization CNN, feature maps are obtained using convolution layers, ReLu layers, and pooling layers. In this article, the Zeiler and Fergus (ZF) model [42] and the visual geometry group (VGG) model [43] are used to generate feature maps. Object proposal regions are generated and transferred from the RPN to the object classification and accurate localization CNN. The object classification and accurate localization CNN combines the object proposal regions with the last feature map to obtain feature vectors of object proposal regions. The feature vectors of object proposal regions are transmitted to the ROI pooling layer to obtain the specified size feature vectors. The specified sized feature vectors are transferred to the full-connected (FC) layer to enable classification and accurate localization of object proposal regions. In the object classification and accurate localization CNN, n is the number of object classes and background. The FC output of object classification is n, and the FC output of object localization is $4n$ representing the coordinates of an object proposal region. The multitask loss function $L(p, k^*, t, t^*)$ of training the object classification and accurate localization CNN is calculated as follows:

$$L(p, k^*, t, t^*) = L'_{\text{cls}}(p) + [k^* \geq 1] L'_{\text{reg}}(t, t^*) \tag{6}$$

$$L'_{\text{cls}}(p) = \sum_i -ln(p_{ik^*}) \tag{7}$$

$$L'_{\text{reg}}(t, t^*) = \sum_{i\in\{x,y,x,h\}} \text{smooth}_{L_1}\left(t_i - t_i^*\right) \tag{8}$$

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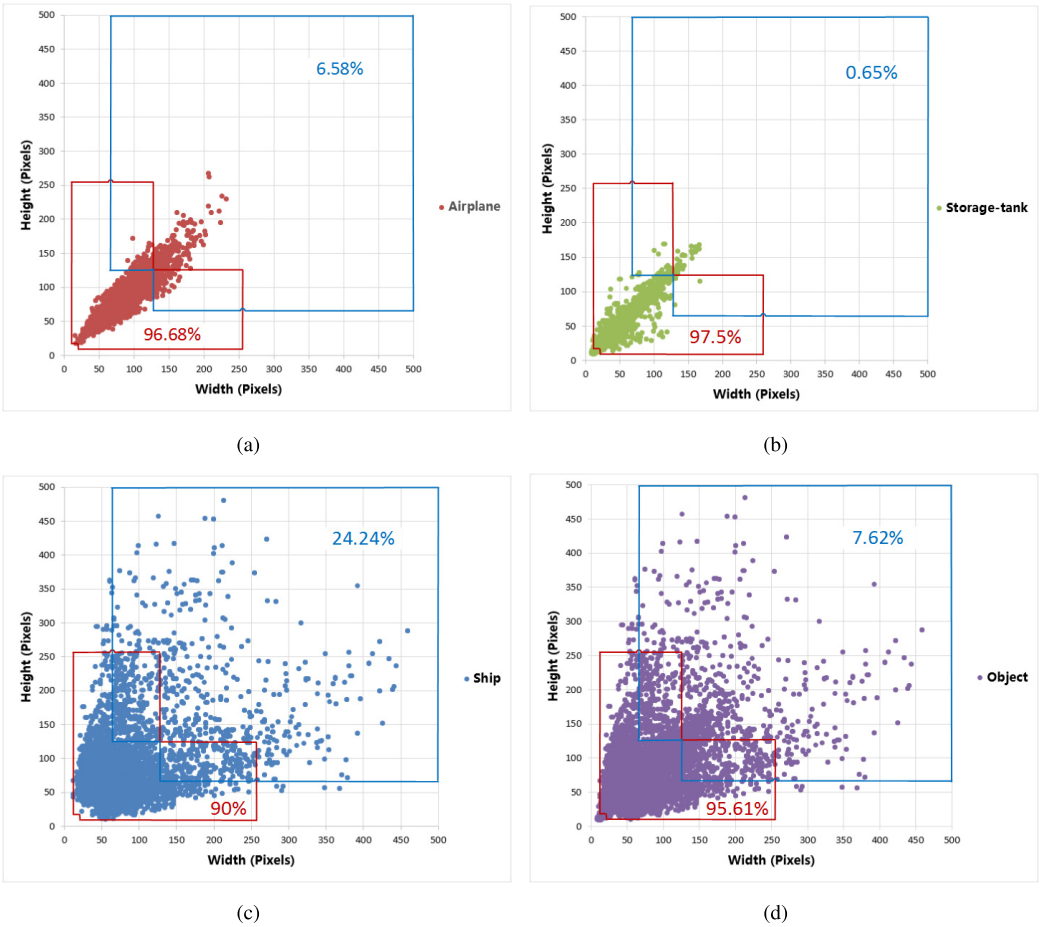


Fig. 5. Statistical results for (a) airplane scales in WHU-RSONE, (b) storage-tank scales, (c) ship scales, and (d) all object scales.

where i is the index of an object proposal region in a mini-batch, and k^* is a true class label of the object proposal region i. If the ground-truth box associated with object proposal region i is not background, $k^* \geq 1$; otherwise, $k^* = 0$. The classification loss L'_{cls} is the standard cross-entropy loss. The Iverson bracket indicator function $[k^* \geq 1]$ evaluates to 1 when $k^* \geq 1$ and 0 otherwise. t_i is a vector representing the four parameterized coordinates of the predicted object proposal region and is $[x_i, y_i, w_i, h_i]^T$. t_i^* is a vector representing the four parameterized coordinates of the ground-truth box associated with object proposal region i and is $[x_i^*, y_i^*, w_i^*, h_i^*]^T$. The regression loss L'_{reg} is the smooth L_1 loss function.

3) Training and Testing of CNN-SOSF: CNN-SOSF is written in Python and C++ (Caffe) [44]. In the training phase, CNN-SOSF is initialized using the pretrained network parameters from ImageNet [45]. CNN-SOSF is trained end-to-end by backpropagation and stochastic gradient descent (SGD) [46]. The loss of CNN-SOSF is obtained by adding the loss of RPN and the object classification and accurate localization CNN. Each mini-batch of RPN arises from an HSRI containing many positive and negative anchors. In an HSRI, 256 anchors are randomly sampled to compute the loss function of a mini-batch of RPN, in which the ratio of the number of positive and negative anchors is close to 1:1. If the number of positive anchors is less than 128 in an HSRI, negative anchors are padded in the mini-batch. Object proposal regions are generated in RPN. Some object proposal regions overlap greatly with each other. To reduce redundancy, NMS is applied to the proposal regions based on their object probability scores. The IoU threshold for NMS is set to 0.7. After NMS, the top 2000 regions for each HSRI are selected for detection. The loss function of a mini-batch of the object classification and accurate localization CNN is computed using 2000 regions. In the experiment, a detection is considered correct if the IoU of its bounding box and ground-truth bounding box is greater than or equal to 0.5; otherwise, it is a false positive. In the training process, a learning rate of 0.001 for 50 000 mini-batches and 0.0001 for the next 25 000 mini-batches is used on WHU-RSONE. The momentum and weight decay are set to 0.9 and 0.0005, respectively.

In the testing phase, an image is input into CNN-SOSF, and 6000 object proposal regions are generated using RPN. We adopt NMS on the 6000 proposal regions based on their object probability scores. The IoU threshold for NMS is set to 0.7. After NMS, we select the top 300 regions per image for detection using the object classification and accurate localization CNN.

III. Experimental Results and Discussion

Large-scale learning samples are the basis of supporting deep learning to achieve high performance. Due to the lack of

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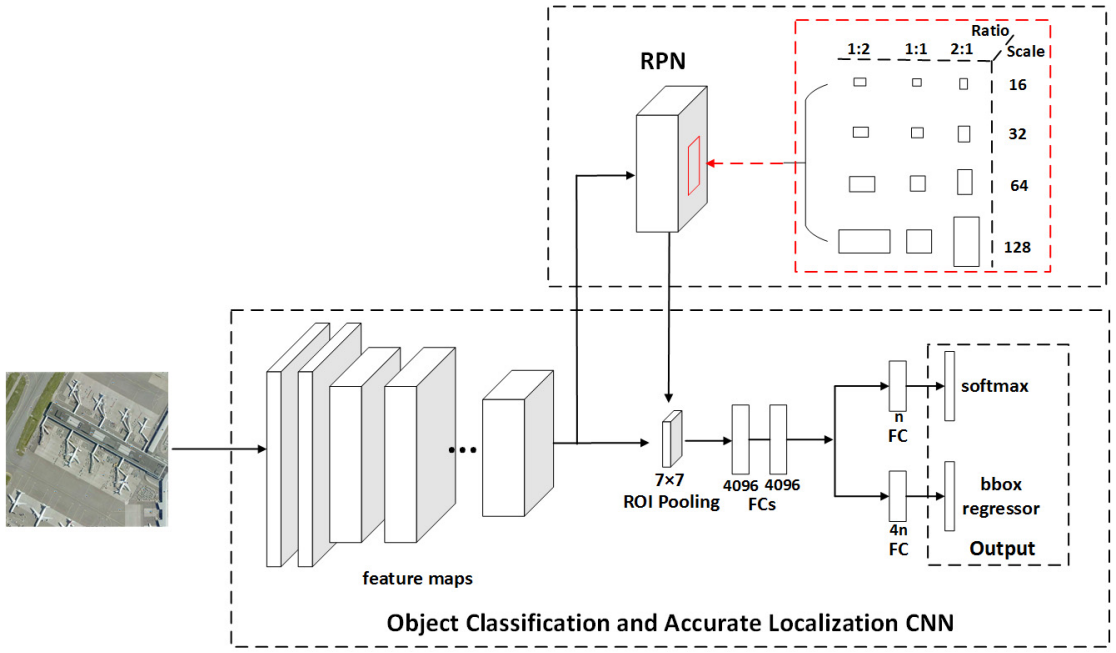


Fig. 6. CNN-SOSF framework.

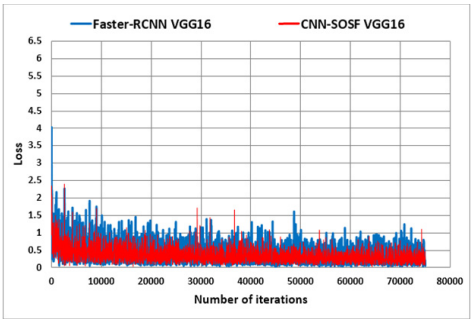
public data sets for object detection in HSRIs [30], an HSRI object detection data set, WHU-RSONE, is established in this article. HSRIs in WHU-RSONE were cropped from Google Earth, Tianditu, and SuperView1 images. There are 5977 HSRIs in WHU-RSONE, and the size of images ranges from 600 × 600 pixels to 1372 × 1024 pixels. The spatial resolution of images ranges from 0.5 to 0.8 m. All images are three-band 8-bit JPEG images. The number of objects in WHU-RSONE is shown in Table I. We make WHU-RSONE publicly available to support the development of related studies.

To examine the effectiveness of CNN-SOSF, qualitative and quantitative comparisons between Faster-RCNN and CNN-SOSF were performed and are described in this article. The ZF and VGG models are used to generate the feature maps of Faster-RCNN and CNN-SOSF, respectively. Experiments are implemented using Python and C++ (Caffe) and performed on a computer with Intel Xeon CPU E3-1220 v5 @3.00 GHz, NVIDIA GTX-1060 GPU, and 16-GB RAM.

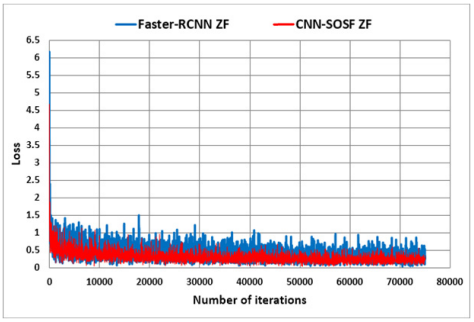
A. Training Loss

In WHU-RSONE, 4781 images are randomly selected as the training set, 598 images are used as the validation set, and 598 images are used as the test set. In the training phase, Faster-RCNN and CNN-SOSF are initialized using the pretrained network parameters from ImageNet. Fig. 7(a) shows the training loss diagram of Faster-RCNN and CNN-SOSF based on the ZF model. Fig. 7(b) shows the training loss diagram of Faster-RCNN and CNN-SOSF based on the VGG16 model.

In Fig. 7, the training loss values of CNN-SOSF based on the ZF and VGG16 models are smaller than those of Faster-RCNN based on the ZF and VGG16 models, respectively. The experimental results show that for HSRIs,



(a)



(b)

Fig. 7. Training loss diagram of Faster-RCNN and CNN-SOSF based on the (a) ZF model and (b) VGG16 model.

CNN-SOSF can better fit the scale of objects and obtain better object detection training model than Faster-RCNN.

B. Mean Average Precision

In this article, mean average precision (mAP) [47] is used as the criteria for object detection. The larger the mAP value, the greater the object detection accuracy of the framework,

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TABLE II

PERFORMANCE COMPARISONS OF FASTER-RCNN AND CNN-SOSF BASED ON THE ZF AND VGG16 MODELS IN TERMS OF AP VALUES AND CONSUMING TIME. THE HIGHLIGHTED NUMBER REFERENCES THE MAXIMUM VALUE IN EACH COLUMN

	Airplane	Storage-tank	Ship	mAP	Time / Per image (s)
Faster-RCNN ZF	0.9261	0.5355	0.6985	0.7200	0.286
Faster-RCNN VGG16	0.9323	0.5240	0.7072	0.7211	**0.318**
CNN-SOSF ZF	0.9403	**0.7181**	**0.7434**	0.8006	0.281
CNN-SOSF VGG16	**0.9542**	0.7071	0.7428	**0.8014**	0.316

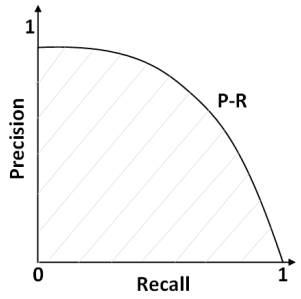


Fig. 8. PRC diagram.

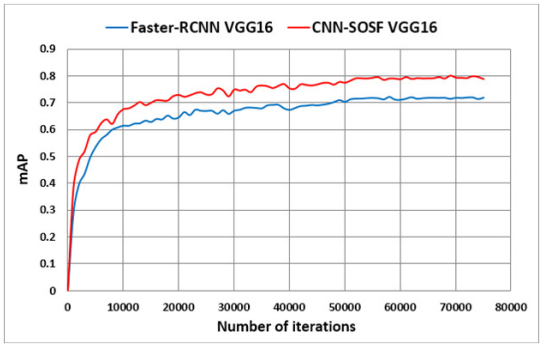
and vice versa. The mAP is calculated as follows:

$$\text{mAP} = \frac{1}{n}\sum_{i}^{n}\text{AP}_i \tag{9}$$

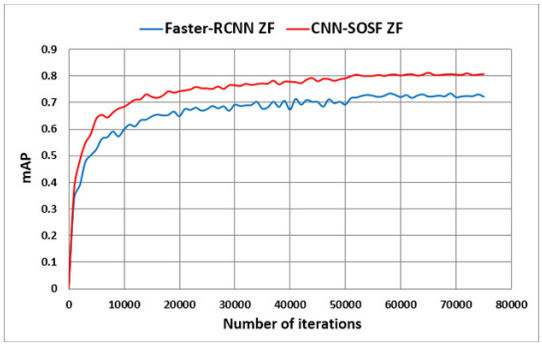
where n is the number of object classes and i is the label of a class. AP_i is the average precision of label i class and is the area under the precision–recall curve (PRC), as shown in Fig. 8.

Fig. 9(a) shows the mAP trend of Faster-RCNN and CNN-SOSF based on the ZF model as the number of training iterations increases. Fig. 9(b) shows the mAP trend of Faster-RCNN and CNN-SOSF based on the VGG16 model as the number of training iterations increases. In Fig. 9, with an increase in the number of training iterations, the mAP values of CNN-SOSF based on the ZF and VGG16 models are larger than those of Faster-RCNN based on the ZF and VGG16 models, respectively. The experimental results show that CNN-SOSF can attain better object detection accuracy than Faster-RCNN for HSRIs.

Table II shows the quantitative comparison results of Faster-RCNN and CNN-SOSF based on the ZF model and on the VGG16 model measured by AP values when the mAP values of Faster-RCNN and CNN-SOSF based on the ZF and VGG16 models are greatest in their respective mAP trend diagrams. In Table II, the AP values of CNN-SOSF ZF for airplanes, storage tanks, and ships are greater than those of Faster-RCNN ZF. The AP values of CNN-SOSF VGG16 for airplanes, storage tanks, and ships are greater than those of Faster-RCNN VGG16. In particular, the AP values of Faster-RCNN ZF and Faster-RCNN VGG16 for storage tanks are 0.5355 and 0.524, respectively. The AP values of CNN-SOSF ZF and CNN-SOSF VGG16 for storage tanks are 0.7181 and 0.7071, respectively. The AP values of CNN-SOSF ZF and CNN-SOSF VGG16 for storage tanks are 18.26% and 18.31% greater than those of Faster-RCNN ZF and Faster-RCNN VGG16, respectively. We obtain significant



(a)



(b)

Fig. 9. mAP trend diagram of Faster-RCNN and CNN-SOSF based on the (a) ZF model and (b) VGG16 model.

performance improvement for storage tanks. Furthermore, the mAP values of Faster-RCNN ZF and Faster-RCNN VGG16 are 0.72 and 0.7211, respectively. The mAP values of CNN-SOSF ZF and CNN-SOSF VGG16 are 0.8006 and 0.8014, respectively. The mAP values of CNN-SOSF ZF and CNN-SOSF VGG16 are 8.06% and 8.03% greater than those of Faster-RCNN ZF and Faster-RCNN VGG16, respectively. Those show that CNN-SOSF can obtain better object detection results than Faster-RCNN in HSRIs.

Table II shows the average consuming time of object detection for per image using four methods in the test set. In Table II, it is efficient for object detection using four methods, and object detection using CNN-SOSF takes less time than Faster-RCNN.

Fig. 10 shows the PRCs of Faster-RCNN ZF, Faster-RCNN VGG16, CNN-SOSF ZF, and CNN-SOSF VGG16. As seen, for airplanes, storage tanks, and ships, the AP values of CNN-SOSF ZF and CNN-SOSF VGG16 are greater than those of Faster-RCNN ZF and Faster-RCNN VGG16. This can be easily explained because the anchor scales of

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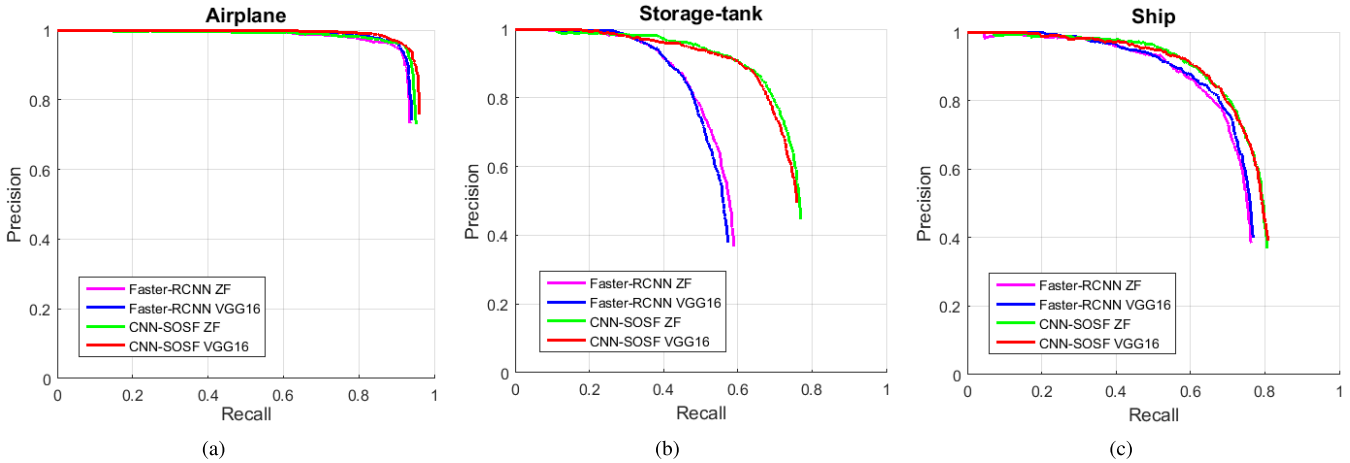


Fig. 10. PRCs of Faster-RCNN and CNN-SOSF based on the ZF and VGG16 models for (a) airplanes, (b) storage tanks, and (c) ships.

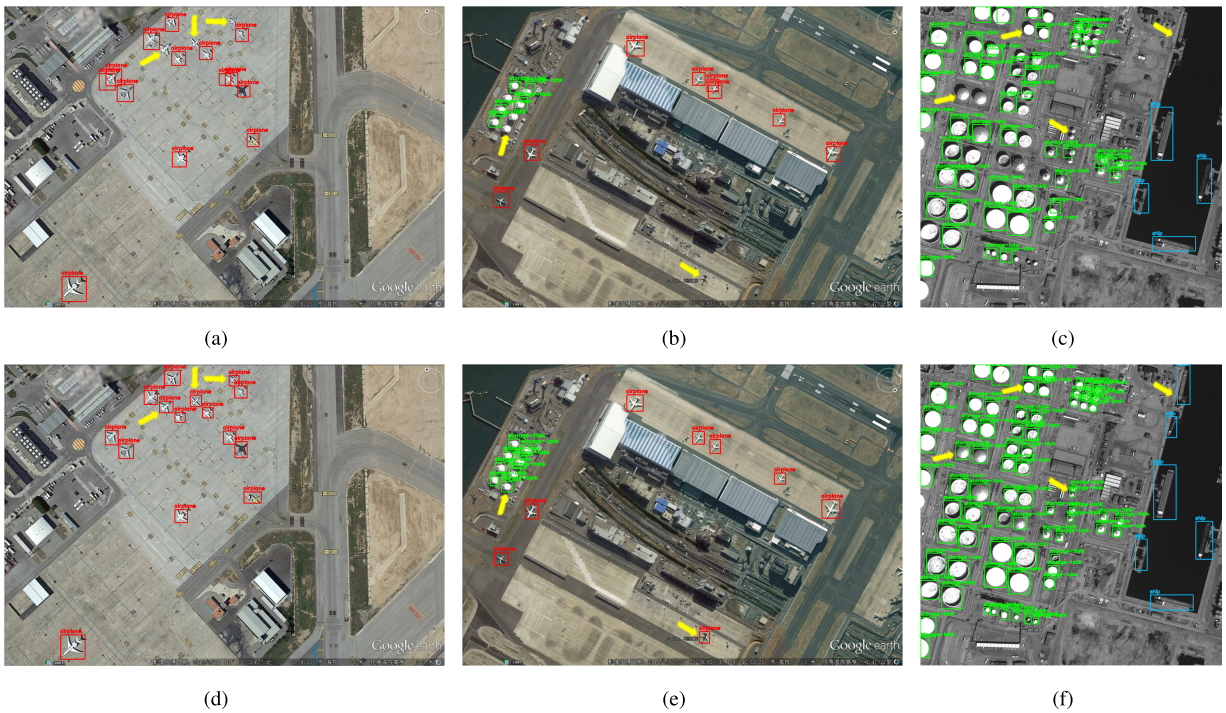


Fig. 11. (a)–(c) Examples of object detection for Faster-RCNN VGG16. (d)–(f) Examples of object detection for CNN-SOSF VGG16.

CNN-SOSF are more suitable for HSRI objects than those of Faster-RCNN. Better object detection results are obtained using CNN-SOSF.

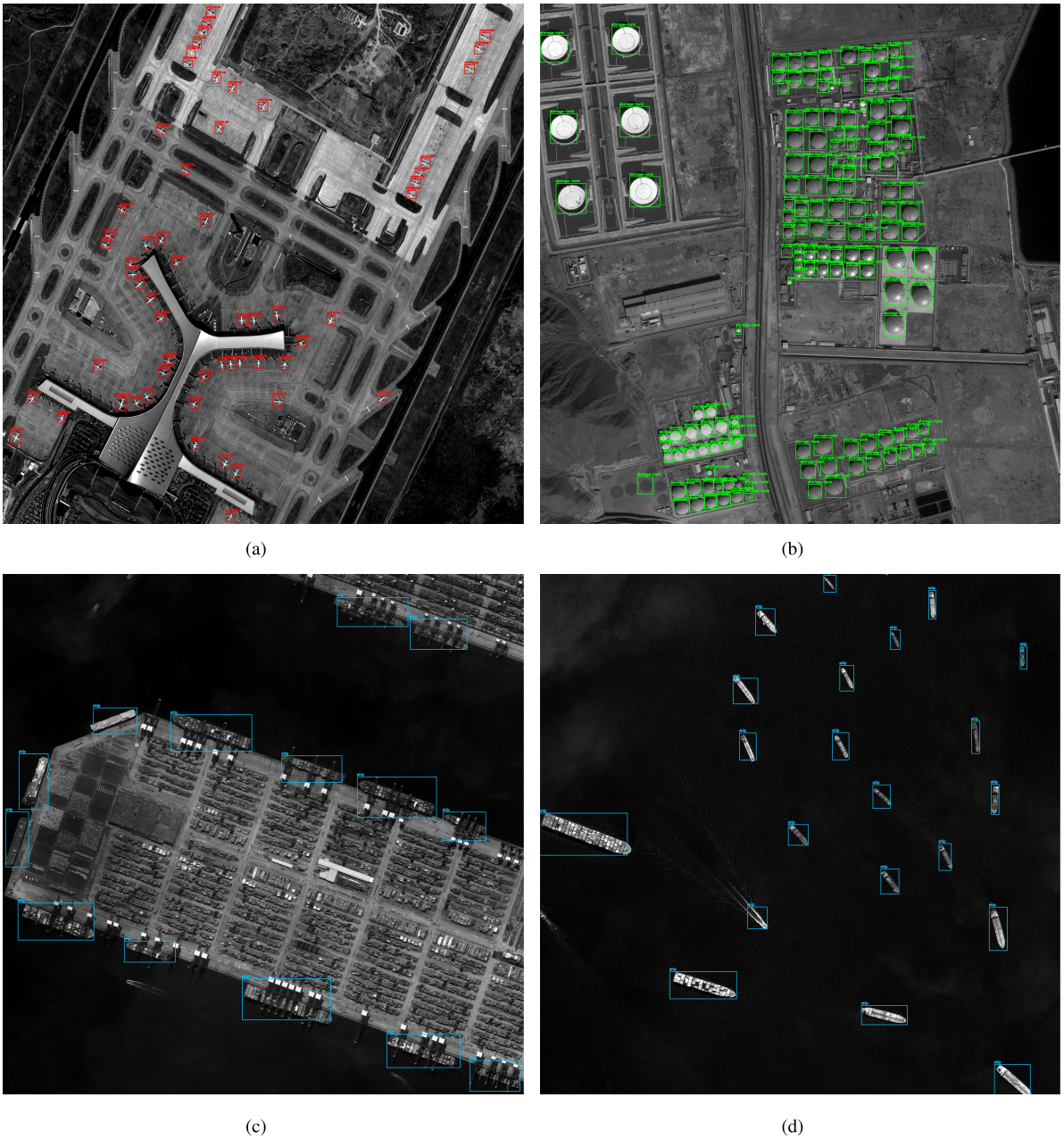
C. Visual Evaluation

In Table II, the mAP value of Faster-RCNN VGG16 is greater than that of Faster-RCNN ZF. The mAP value of CNN-SOSF VGG16 is greater than that of CNN-SOSF ZF. Therefore, Faster-RCNN VGG16 and CNN-SOSF VGG16 with greater detection accuracy are evaluated by visual comparison. The confidence threshold of object detection for Faster-RCNN VGG16 and CNN-SOSF VGG16 is set to 0.8. Fig. 11(a)–(c) shows the examples of object detection for Faster-RCNN VGG16 in the WHU-RSONE test set. Fig. 11(d)–(f) shows the examples of object detection for CNN-SOSF VGG16 in the WHU-RSONE test set.

In Fig. 11, for objects (airplanes, storage tanks, and ships) with smaller scales as indicated by the yellow arrows, Faster-RCNN VGG16 fails to detect, and CNN-SOSF VGG16 can detect them well. The experimental results show that CNN-SOSF VGG16 is more suitable for small-scale HSRI objects and can obtain better object detection results than Faster-RCNN VGG16.

To further verify the effectiveness of the CNN-SOSF framework for HSRI object detection, CNN-SOSF VGG16 is applied to object detection in GaoFen2 (GF2) panchromatic images. The spatial resolution of GF2 panchromatic images is 0.8 m. The four GF2 panchromatic images are 2500 × 2500 pixels and their object detection results using CNN-SOSF VGG16 are shown in Fig. 12. Since the size of four GF2 panchromatic images is large for CNN-SOSF VGG16 input, images are divided into several contiguous

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(a) (b) (c) (d)

Fig. 12. (a)–(d) Object detection results of GF2 panchromatic images.

image blocks for object detection whose size is 800 × 800 pixels. During the division into blocks, we set an overlap of 400 pixels for each contiguous image block. After each image block is detected, the detection results of each image block are merged into the original image. In the merger process, we adopt NMS on the bounding boxes based on their scores to eliminate redundant bounding boxes. The IoU threshold for NMS is set to 0.1.

In Fig. 12, airplanes, storage tanks, shore ships, and offshore ships in GF2 panchromatic images can be accurately detected using CNN-SOSF VGG16. The experimental results show that CNN-SOSF VGG16 can be effectively applied for object detection in GF2 images, and the CNN-SOSF framework has good robustness and universality in the HSRI object detection.

IV. Conclusion

The ROI scale of object detection and the object feature representation are two vital factors in the HSRI object detection. With respect to these two issues, a novel HSRI object detection method based on CNN-SOSF is proposed in this article.

For obtaining the suitable ROI scale of object detection, an HSRI object detection data set, WHU-RSONE, is defined and discussed in this article. There are 51 866 geospatial objects (15 703 airplanes, 24 692 storage tanks, and 11 471 ships) in WHU-RSONE. WHU-RSONE can be downloaded at https://pan.baidu.com/s/1yQD9ZoP-lWY1bfvSYpG0KQ. The scales of objects in WHU-RSONE are counted, and statistical results show that anchors obtained using RPN with four scales (16 pixels, 32 pixels, 64 pixels,

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and 128 pixels) and three aspect ratios (1:1, 1:2, and 2:1) can fit ROI scales of object detection well in HSRIs.

The CNN-SOSF framework for object detection in HSRIs is designed using a suitable ROI scale of object detection. The object features obtained using CNN-SOSF have good universality and robustness. Faster-RCNN and CNN-SOSF based on the ZF and VGG16 models are compared and analyzed using WHU-RSONE. CNN-SOSF outperforms Faster-RCNN and obtains better object detection results. The CNN-SOSF framework can effectively be applied in object detection in GF2 images and demonstrates good robustness for HSRI object detection. In further work, we will add the object directly to the CNN-SOSF framework and improve its detection performance.

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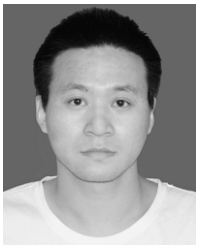
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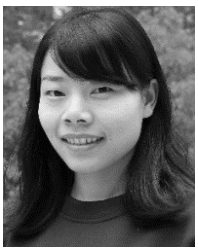
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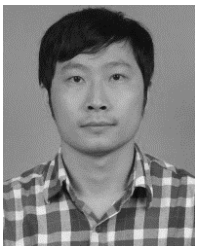
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